

Smart Energy Consumption Prediction

Forecasting Daily Household Electricity Usage and Flagging High-Usage Days

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Abstract

This project forecasts daily household electricity consumption and identifies days likely to see unusually high usage, so that a household or energy provider can plan simple load-shifting responses in advance. Using the UCI Individual Household Electric Power Consumption dataset combined with historical weather data, an XGBoost regression model was trained on calendar, lag, rolling-average, and weather-derived features, and tuned with time-aware cross-validation to avoid lookahead leakage. The final model achieved a mean absolute error of 247.80 (versus 286.29 for a naive "same as yesterday" baseline), an improvement of roughly 13-17% over simple baselines. Beyond the headline accuracy number, the project places equal weight on understanding model behaviour: SHAP analysis confirmed that recent consumption history is the dominant driver of predictions, weekends run about 17% higher than weekdays, and a purpose-built "heating degree" feature captures a real, physically sensible weather effect. Investigation of the largest prediction errors traced most of them to occupancy anomalies - travel or guests - that no feature in the dataset could have anticipated, which is reported here as an honest limitation rather than smoothed over. A threshold-based alert system for high-usage days was built and evaluated on a precision-recall basis, favouring recall (catching real spikes) over precision, an appropriate trade-off for a low-stakes advisory tool.

1. Problem Statement

Energy providers and households both benefit from knowing in advance when consumption is likely to spike. Providers use this information to balance supply and demand more efficiently; households can use it to shift discretionary, high-load activities (laundry, dishwashing, water heating) to off-peak periods. Day-to-day electricity demand is difficult to predict because it depends on a mix of routine behaviour, calendar effects, and weather, several of which interact non-linearly. The goal of this project is to build a forecasting model that is accurate enough to be useful, and to understand - rather than simply report - where and why it fails.

2. Dataset

- Power consumption: UCI Individual Household Electric Power Consumption dataset - minute-level readings from a single household near Paris, December 2006 to November 2010.
- Weather: historical daily temperature for the same location and period, pulled from Open-Meteo.

3. Approach

- Cleaned the raw minute-level readings and resampled them to daily totals.
- Engineered features: calendar information (day of week, month, day of year), lag features (previous day, previous week), rolling averages and rolling standard deviation, and a "heating degree" feature capturing distance below a comfortable temperature.
- Trained an XGBoost regression model and tuned hyperparameters with RandomizedSearchCV, using time-aware cross-validation so the model is never evaluated on data that precedes its training window.
- Benchmarked the tuned model against naive baselines ("same as yesterday", "same as last week") rather than judging accuracy in isolation.

- Used SHAP (SHapley Additive exPlanations) to understand which features the model actually relies on, and how.
- Investigated the days with the largest prediction errors against raw sub-metering data (kitchen, laundry, water heater) to understand root causes.
- Built and validated a threshold-based recommendation layer that flags predicted high-usage days.

4. Results

The tuned XGBoost model was evaluated against two naive baselines on a held-out time period. Results are summarised below.

Model	MAE	RMSE
Naive — same as yesterday	286.29	405.35
Naive — same as last week	387.99	532.19
Tuned XGBoost (this project)	247.80	338.54

The tuned model improves on the best naive baseline by roughly 13-17% on both MAE and RMSE - a real but modest improvement, consistent with the fact that recent consumption is already a strong predictor of near-term usage.

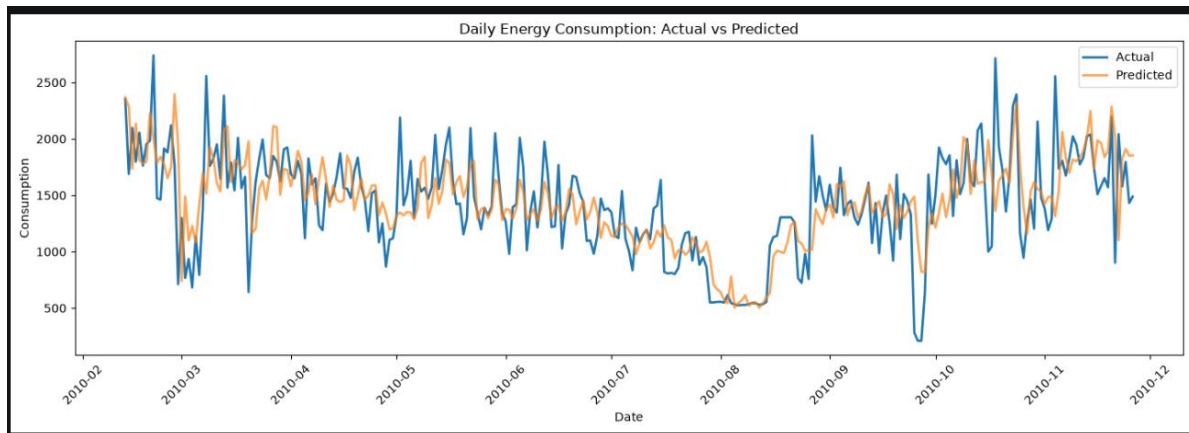


Figure 1. Actual vs. predicted daily energy consumption over the evaluation period.

4.1 Key Findings

- Recent consumption matters most: yesterday's value and the 7-day rolling average dominate the model's predictions.
- Weekends are genuinely different: consumption runs about 17% higher on Saturdays and Sundays than on weekdays.
- Weather helps, but less than expected: using raw temperature directly slightly hurt accuracy. Engineering a "heating degree" feature (distance below a comfortable temperature) corrected this and produced a physically sensible effect in the SHAP plots - cold days push predictions up - though it remains a smaller effect than recent usage history.

4.2 Feature Importance (SHAP)

SHAP values were used to move beyond a simple feature-importance ranking and understand the direction and consistency of each feature's effect on individual predictions.

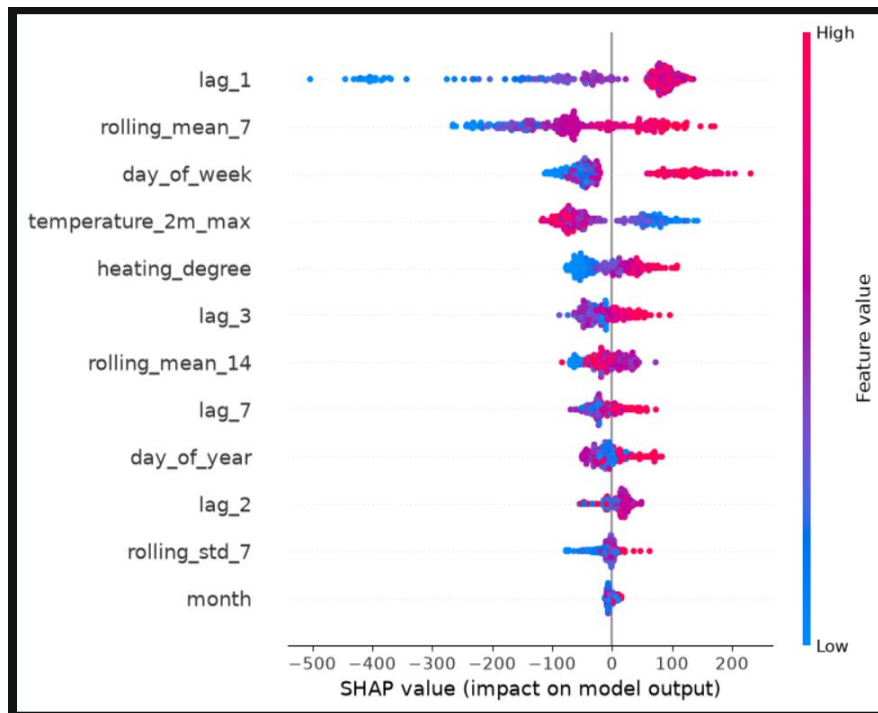


Figure 2. SHAP summary plot showing each feature's impact on model output, ranked by importance.

The plot confirms that lag_1 (yesterday's consumption) and rolling_mean_7 are the two strongest drivers, followed by day_of_week and the temperature-related features. The colour gradient shows that high recent consumption (red, high lag_1) consistently pushes predictions upward, while low recent consumption pushes them downward - the relationship is monotonic and behaves as expected, which is a useful sanity check on the model.

5. Error Analysis

Rather than stopping at an aggregate accuracy score, the residuals were examined over time and the ten largest errors were cross-checked against raw sub-metering data.

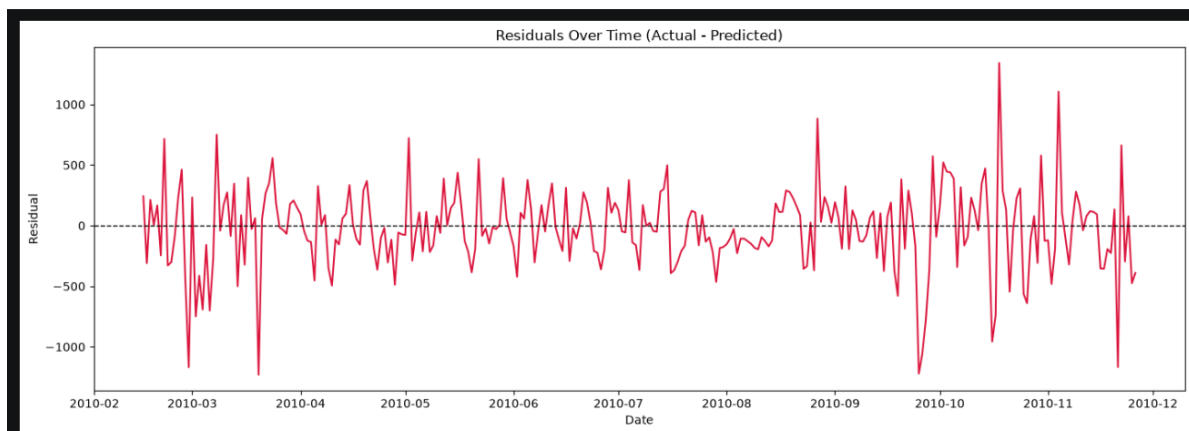


Figure 3. Residuals (actual minus predicted) over the evaluation period.

Residuals are centred near zero with no strong long-term drift, indicating the model is not systematically biased over time, though there are periodic spikes corresponding to specific high-error days.

- 8 of the 10 largest errors traced back to occupancy anomalies - the house being unusually empty (likely travel) or unusually busy (guests, heavy simultaneous appliance use) - confirmed by cross-referencing sub-metering channels.

- None of these occupancy events are observable from any feature available in this dataset, so this represents a genuine limitation of the available data rather than a weakness in the model or feature engineering.

6. High-Usage Day Alerts

True high-usage days are rare - about 2% of all days in this dataset - which makes threshold selection a precision/recall trade-off rather than a single "correct" number. Multiple thresholds were tested explicitly rather than picking one that looked good on a single metric. The chosen threshold (85th percentile of predicted consumption) catches approximately 67% of true high-usage days (recall), at the cost of a relatively low precision of roughly 9% - meaning most flagged days are false alarms. This trade-off was chosen deliberately: for a low-stakes advisory tool, the cost of a missed real spike is higher than the cost of an occasional unnecessary alert.

7. Limitations

- Trained on a single household - will not generalise to other households without retraining.
- No occupancy signal is available (whether people are home, travelling, or hosting), which was found to be the single largest source of prediction error.
- Only about four years of data, so the model has not seen every possible seasonal extreme.
- The high-usage alert system is tuned to favour recall over precision by design - worth noting if reused for a more sensitive application than a household advisory tool.

8. Conclusion

This project shows that a fairly standard gradient-boosted model, combined with careful feature engineering and honest evaluation against baselines, can meaningfully improve on naive forecasts of household electricity consumption. Just as importantly, digging into where the model is wrong - rather than stopping at a single accuracy number - surfaced a concrete, well-evidenced limitation (unobserved occupancy) that no amount of further model tuning could fix, and clarified the real trade-offs involved in turning a forecast into an actionable alert.